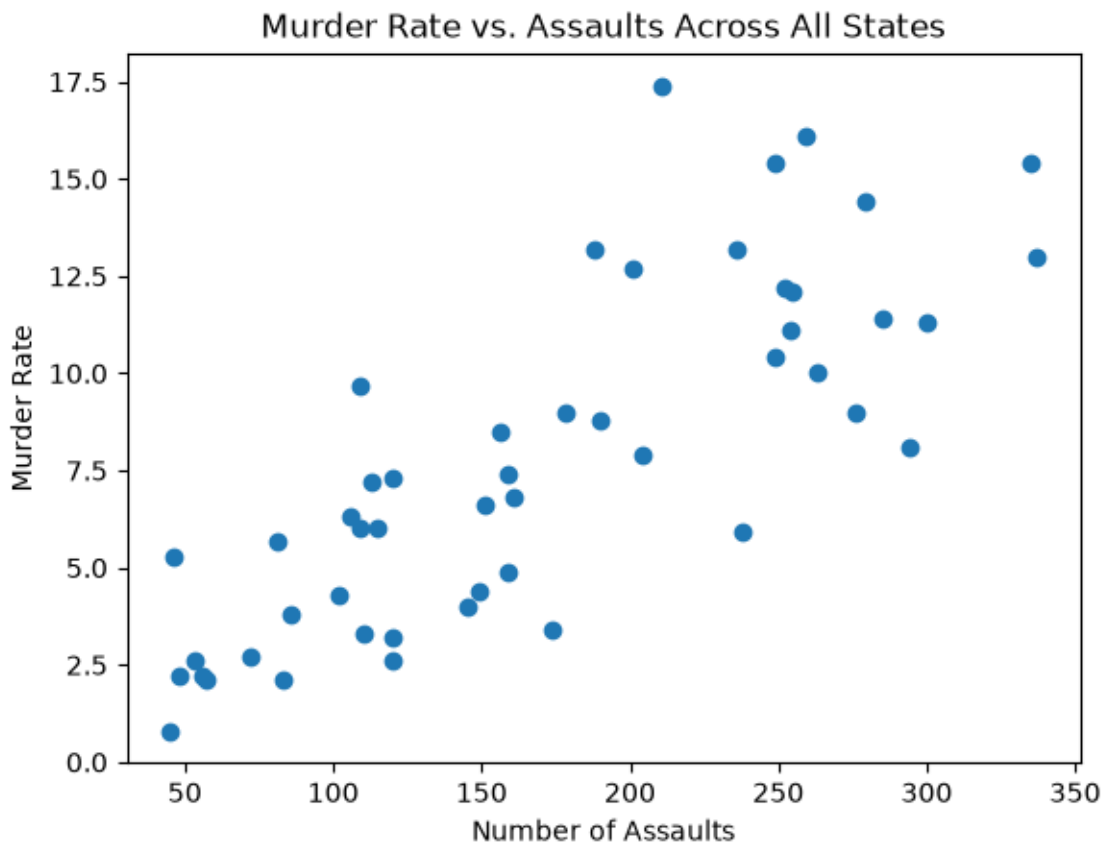
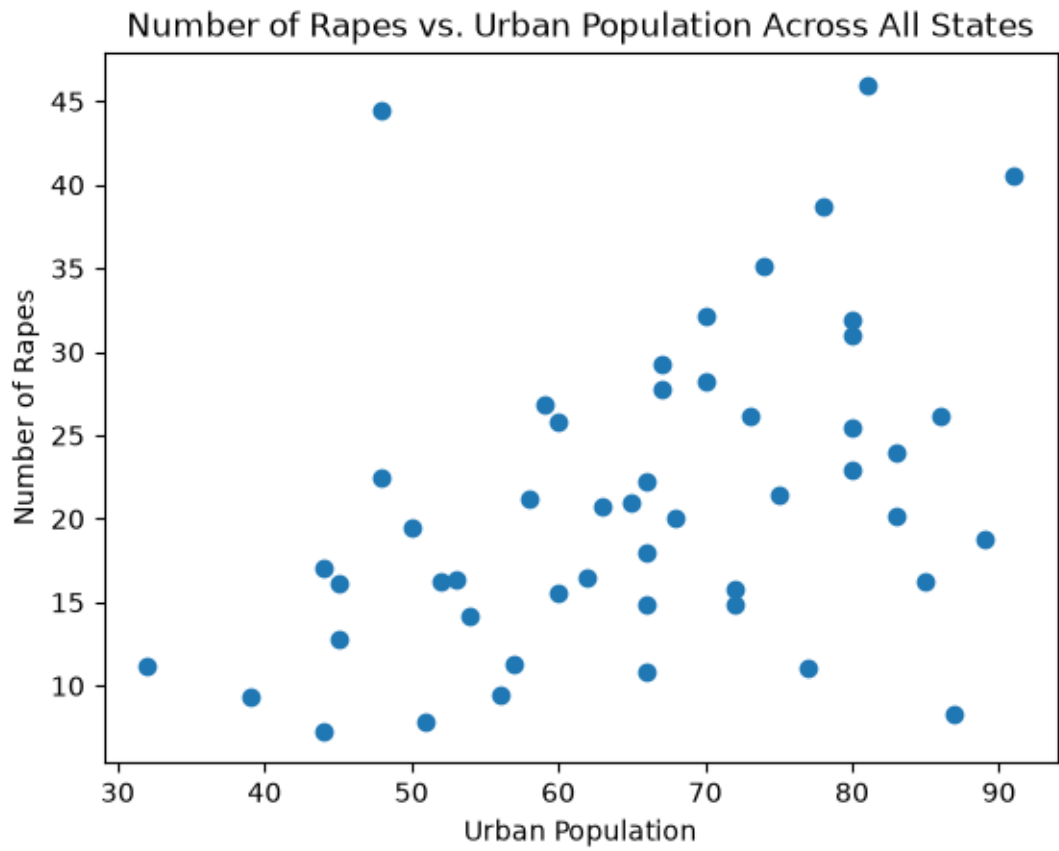


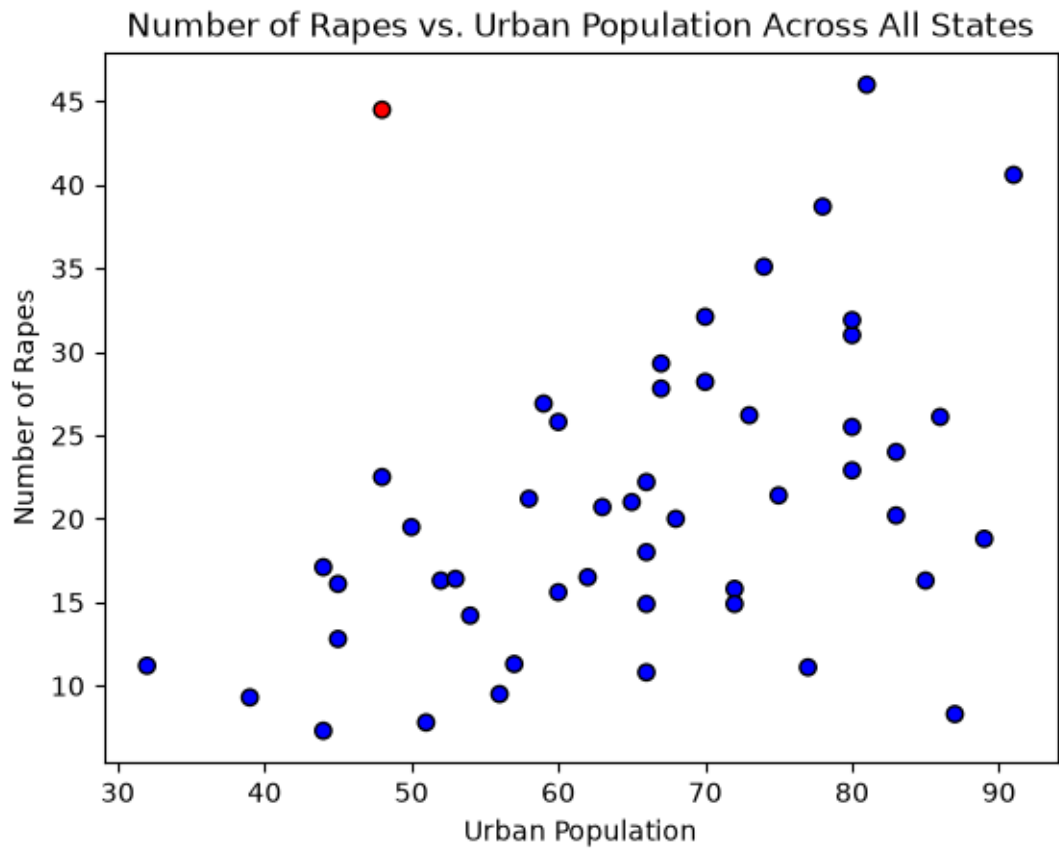
Crime Statistics Across States

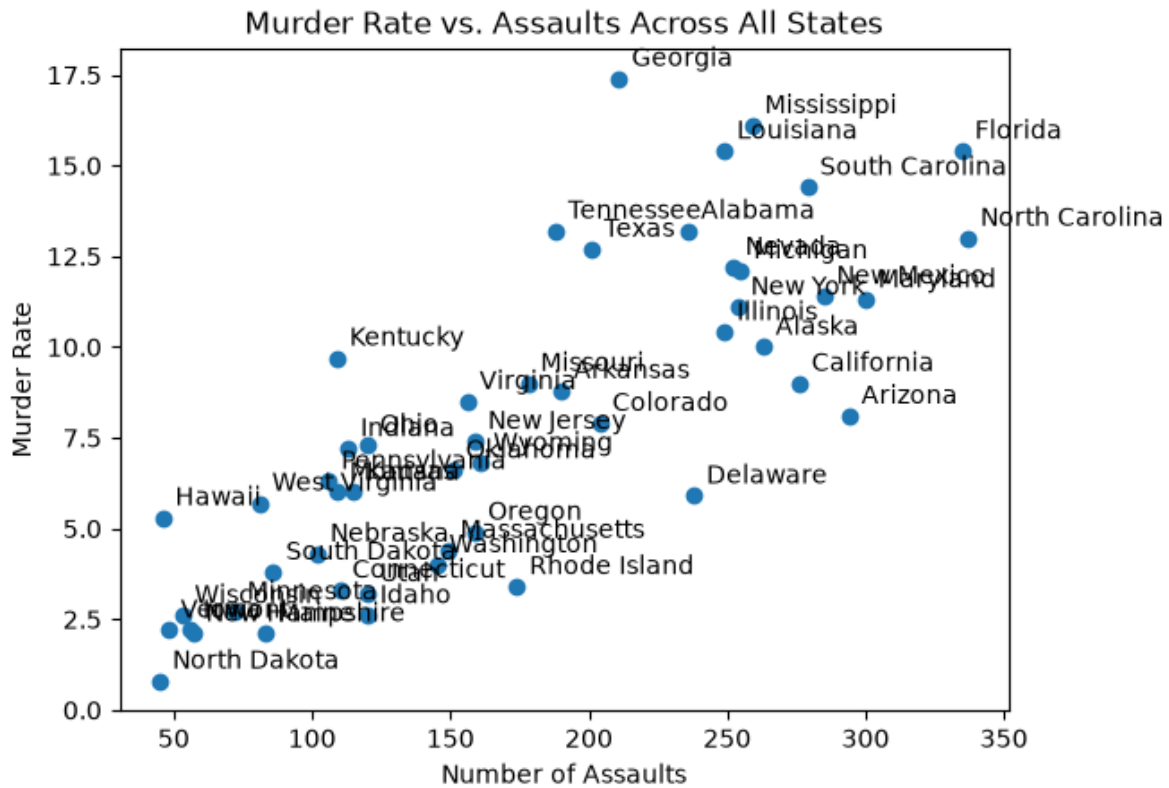
By Jordan Collins

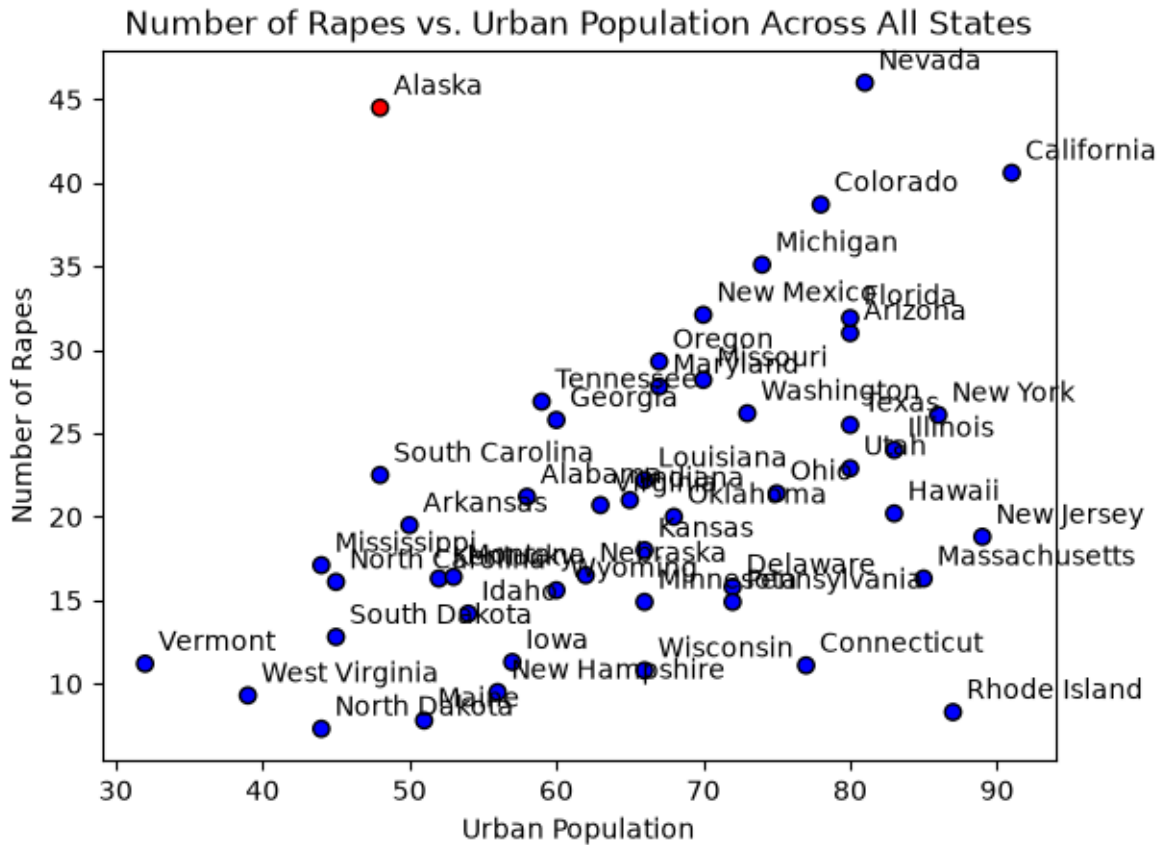


When plotting Murder Rate vs. Number of Assaults Across All States, we see a general trend that indicates the higher number of assaults that occur within a state, the higher the murder rate is likely to be.

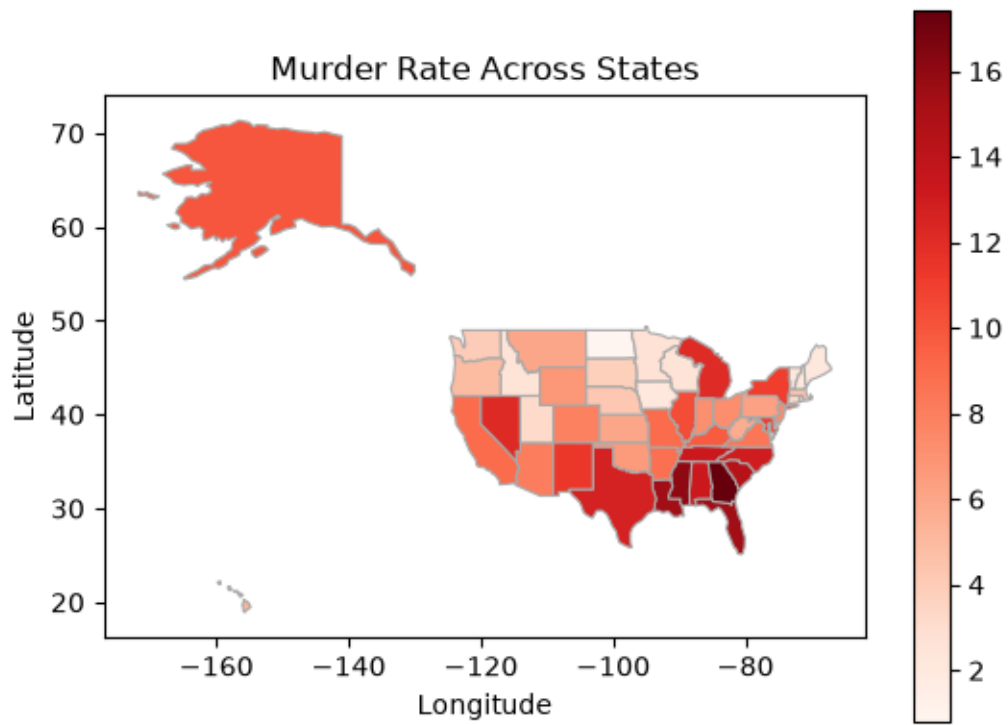


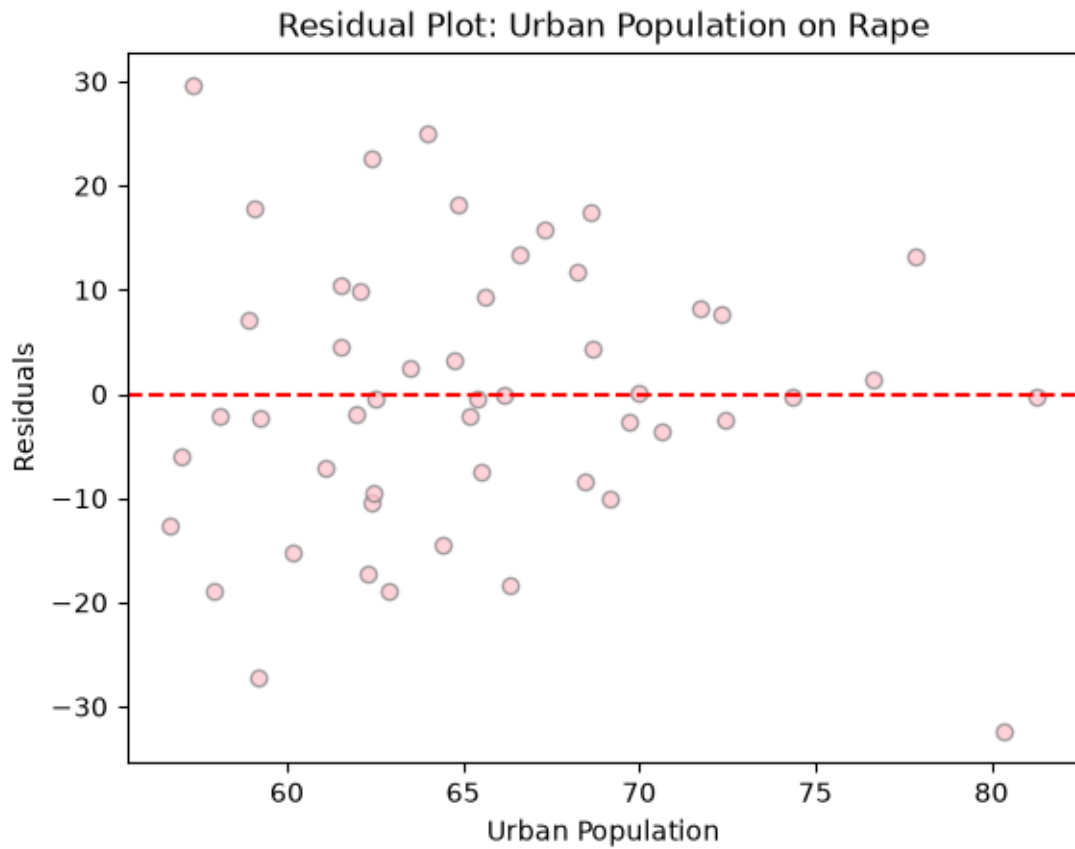




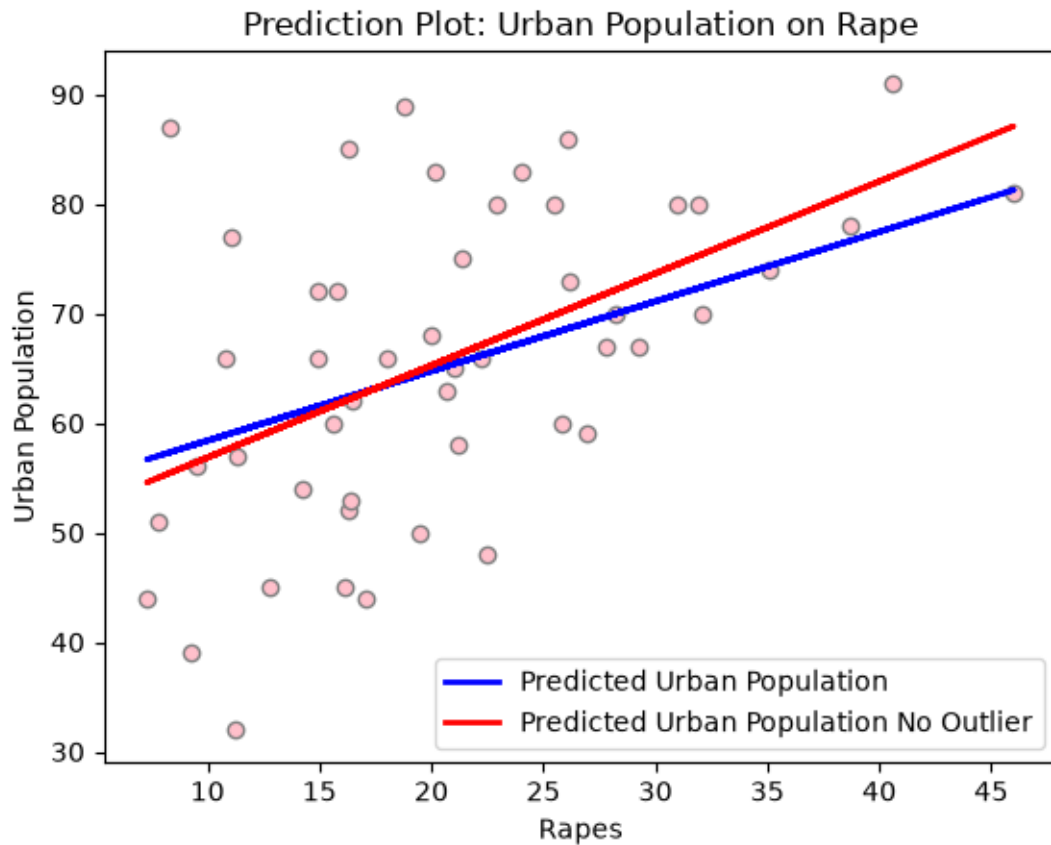


Interestingly, there are many high murder rates and number of assaults in the southern states of the US. There is also a general trend showing that as urban population increases, so do the number of rapes in that state. Alaska is the one state that is an outlier with a relatively smaller urban population, but a high number of rapes.





Funnel shape is seen in this graph, indicating heteroscedasticity. Low urban populations have a wider range of absolute residual values while higher urban populations have lower absolute residual values. There is one outlier in the bottom right.



Removing the outlier did change the results. The red line seems to show a more distinctive trend. However, there is still a large spread of urban populations, especially when there are fewer rape cases in a state. This makes the trendline seem less valuable and true to the data.